Project 3: Classification and AI Methods
Grade 7

Overview

In most AI applications, AI models must make decisions or identify the category of a given input. The project lets students create a tree to make decisions. The classroom activity involves classification (identifying categories) of various types of dal, followed by a project activity based on fruit cards.

Learning Outcomes

- **Classification Algorithm.** Students will create tree diagrams to identify the object based on input diagrams.
- **Efficiency of Algorithms.** There are multiple ways of creating the tree. The classroom activity will let students compare the efficiency of various approaches.
- **Testing of the AI Model.** Students will test their model with fresh inputs.

Material Required

- Various types of dal grains (approx 50 gm each for the whole class)
- Print of fruits cards among a group of 3 or 4



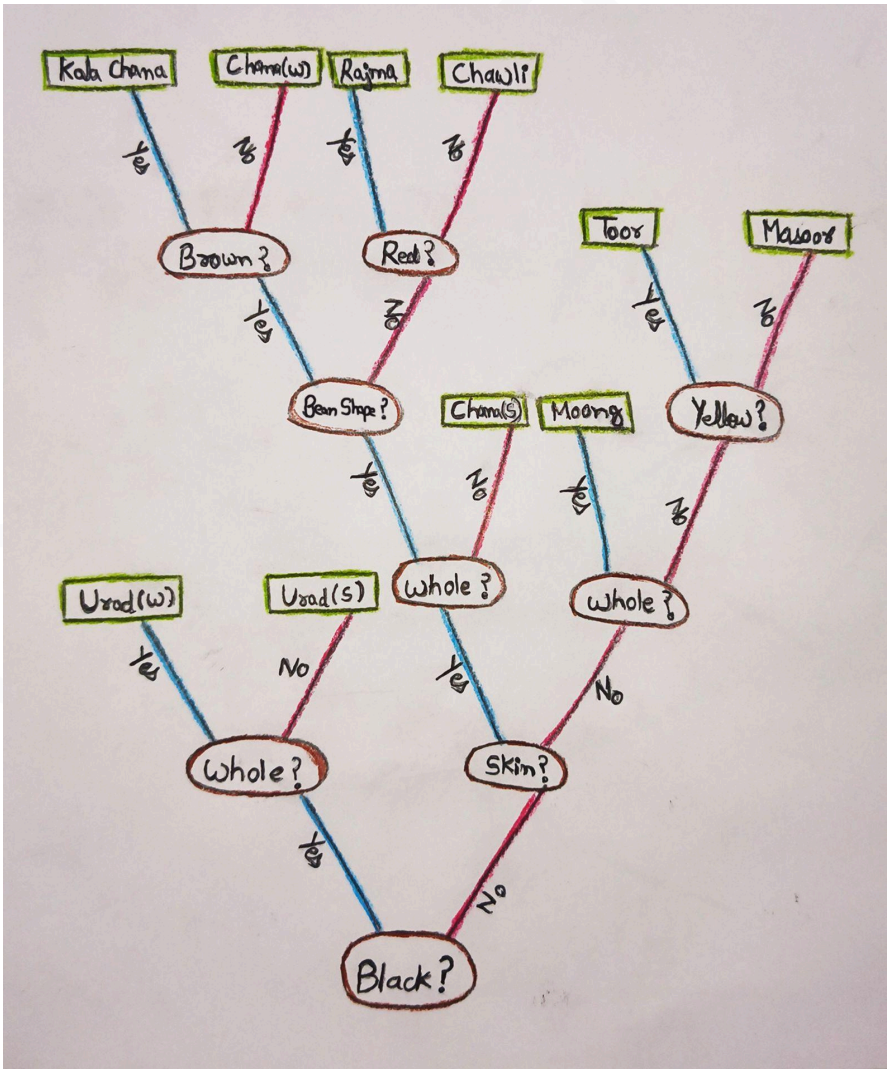
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Project Plan

Project Activity	Time	Description
Launch	10-15 mins	The teacher will ask students to create a tree diagram that will identify the type of dal. Demonstrate a classification tree for 4 types of dal to give students a hint on how to create a tree.
Explore	20-25 mins	Let the students represent their tree diagrams in the classroom. Teachers ask exploring questions based on the analysis of the tree.
Implement	1 week	The students will create a classification model for the type of fruit cards given at the end. Template : https://tinyurl.com/fruitscards

Launch

The activity focuses on the identification of various types of dal by looking at their features, i.e. Whole (W)/split (S) dal, colour of the dal, and presence/absence of skin.



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- Teachers and students will bring various types of dal from their homes. Put the grains into groups on the chart paper, and write their names and features below each group.

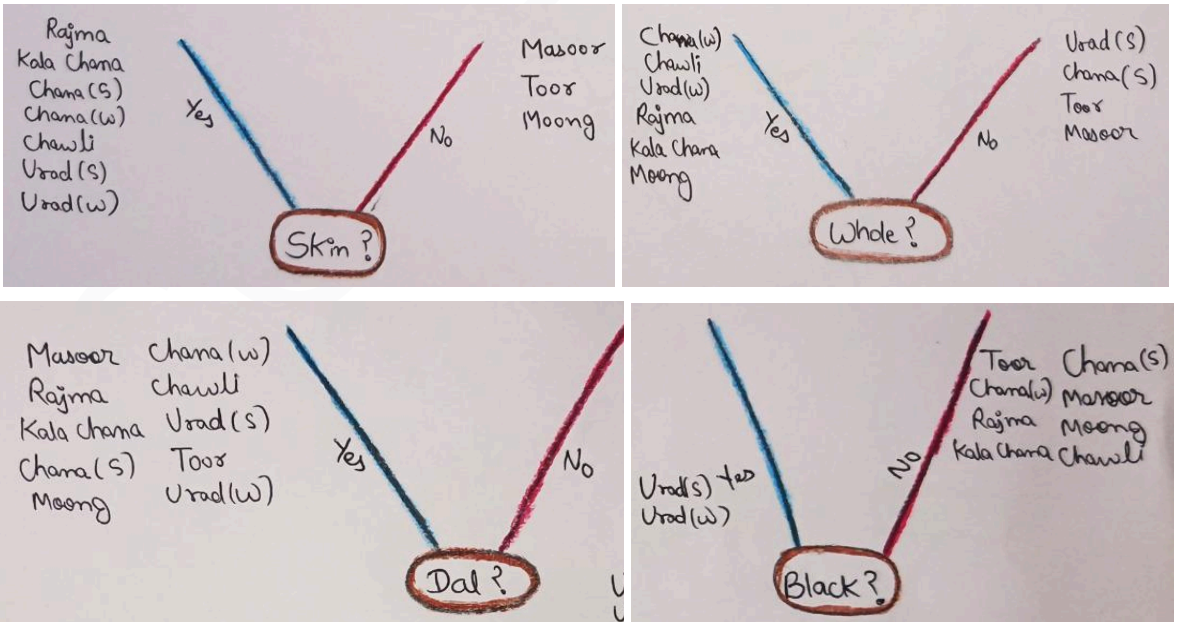
- The students have to create a set of questions to identify the dal based on this feature. The figure shows one such way of finding the category. Create your own set of questions and their order.

- By asking a sequence of yes/no questions based on various features, the tree identifies the category. Start at the bottom. Take one piece of dal, ask the question at the node and place the dal under either the YES or NO branch. When you have done this for all of the various dal, you've divided the set into two groups.

Explore

1. The first question in the decision tree asked whether a dal is black, which split the dal into two parts - one part has 8 types and the other one has 2 types. What would be a better split if we wanted a decision tree with fewer questions?
    a. 8 : 2
    b. 7 : 3
    c. 5 : 5
    d. 10 : 0

2. A 5:5 split, with 5 types in each, could be a good choice because it divides the total number of types into two equal halves and might help us find the category more quickly. For the following scenario, which feature would you ask to split the dals in half?



    a. Skin
    b. Whole/Split
    c. Dal
    d. Colour

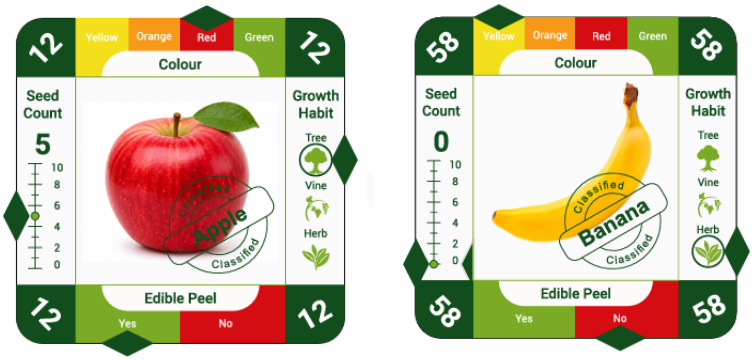
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3. If each question halves the group, how many questions would it take to identify 12 dals?
    a. 2
    b. 3
    c. 4
    d. 11

Implement

Draw a tree that classifies fruits based on features like colour, seed count, growth habit, and peel edibility. There are template data cards for various fruits and their properties. Training fruit data are given that are already classified, each described by properties like:

- Colour: Yellow/Orange/Red/Green
- Growth Habitat: Tree/Vine/Herb
- Edible peel: Yes/No
- Seed Count: Number from 0 - 10



Along with the training cards, testing data cards are given in the template that contains only feature properties, not the images, and students need to identify the fruit based on those properties. The correct category for the test data cards is shown below for the teachers. Numbers on the cards are given just for the references.

Test Card Number	Fruit Category
24	Mango
46	Orange
38	Strawberry
29	Watermelon
62	Banana

Reference: Lehner, Lukas, and Martina Landman. "Unplugged Decision Tree Learning–A Learning Activity for Machine Learning Education in K-12."

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